Final Project Proposal

Graded

Student

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Total Points

5 / 5 pts

Question 1

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5 / 5 pts

✔ **– 0 pts** Correct

– 0.01 pts Not a sequential problem

– 0.01 pts No discussion of source of uncertainty

– 0.01 pts Insufficient detail of the problem

– 2.5 pts More effort needed (see the project proposal instruction on the website for expectations and suggestions)

Strong, well-motivated proposal with clear sequential framing and solid uncertainty modeling. To keep the scope manageable, I'd recommend prioritizing the translational (3-DOF) model and postponing any chance-constrained or 6-DOF extensions unless you make rapid early progress. Good luck!

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Final Project Proposal: POMDP Framework for Fuel Mode Switching in Orbital Rendezvous and Docking

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1. Goal

We aim to develop a decision-making framework for an autonomous chaser spacecraft docking with a satellite, as autonomous servicing and debris-removal missions become more common in orbit. For this task, our satellite will have a dual propulsion system, combining high-thrust chemical engines with low-thrust cold gas thrusters to balance agility and efficiency, commonly seen on many satellites (Scheinman et al., 2019). However, little research exists on optimization of automatically switching between these modes in real time instead of relying on a pre-programmed schedule. (Garrett and Isaacson, 2021).

Our algorithm will allow the spacecraft to dock safely, efficiently manage fuel, and adapt its propulsion-mode decisions in the presence of sensor noise, thrust variation, and timing constraints. Our reward metrics are total fuel consumption, final docking accuracy, and the rate of collision-free trajectories.

2. Decision Making

We set up the problem as a sequential decision-making task: at each time step, the chaser decides on both a thrust vector (magnitude and direction) and a discrete choice of propulsion mode (high vs. low thrust). The spacecraft's state evolves over time according to discrete-time dynamics, and because actions at each step affect all future states, the problem is inherently sequential and cannot be treated as a one-shot time-invariant optimization. The simplified translational model, based on linearized Clohessy–Wiltshire (Hill's) equations, is a well-established standard in orbital rendezvous and proximity operations analyses, including those used in NASA's guidance, navigation, and control studies (Clohessy and Wiltshire, 1960; Center, 2010).

The spacecraft does not have perfect knowledge of its state, so its actions are based on an uncertain belief of its state. Therefore, the problem is modeled as a Partially-Observable Markov Decision Process (POMDP), which is modeled with the following structure:

- **State** (s_t): Relative position (x, y, z), relative velocity $(\dot{x}, \dot{y}, \dot{z})$, remaining fuel, and current propulsion mode.
- **Action** (a_t): Thrust vector (pitch, yaw, and magnitude) and discrete mode selection (chemical or electric propulsion).
- **Transition** ($P(s_{t+1} \mid s_t, a_t)$): Obtained from the discretized Hill's equations with process noise $w_t \sim \mathcal{N}(0, Q)$ representing thrust variability and modeling uncertainty.

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- **Observation** (o_t): Noisy sensor measurements of relative position and velocity,

$$o_t = Hs_t + v_t, \quad v_t \sim \mathcal{N}(0, R),$$

  where H is the observation matrix and R is the observation noise covariance.

- **Reward** ($R(s_t, a_t)$): Penalizes fuel consumption, long time-to-dock, and proximity violations, while rewarding successful docking within a defined safety envelope.

We will also decide whether to incorporate full attitude control (6-DOF) or assume stable attitude and restrict the dynamics to translation only, depending on time and complexity. This would add orientation (q_w, q_x, q_y, q_z), and angular velocity $(\dot{\omega_x}, \dot{\omega_y}, \dot{\omega_z})$ to the state

3. Sources of Uncertainty

1. **Navigation (sensor) uncertainty:** Position and velocity estimates are noisy due to measurement error and geometry (e.g., GPS DOP effects). We will model this via Gaussian observation noise in measurements, with representative values of $\sigma_p \approx 0.05$ m for position and $\sigma_v \approx 0.01$ m/s for velocity, consistent with near-field optical or lidar-based relative navigation systems.

2. **Truth uncertainty:** There are also deviations in quantities not directly measured by the spacecraft. For actuation, the spacecraft's thrust magnitude, thrust direction, and mass flow rates will all have deviations from their expected values as "commanded". Additionally, there is uncertainty in disturbances that are not directly observed by the spacecraft, such as J2, drag, and gravity gradient. For an initial model, these can be combined into a single small disturbance of process noise in the acceleration: $\sigma_a \approx 2 \times 10^{-5}$ m/s^2. For a higher-fidelity model, there could also be multiple separate disturbance distributions for each effect (i.e. one for thrust magnitude, one for drag, etc.).

4. Proposed Plan

We will build a simulation that integrates both translational dynamics (via discretized Hill's equations) and mass calculations (via fuel depletion in each propulsion mode). Sensor measurements will be synthetically generated by consulting sensor datasheets and adding Gaussian noise to the simulated true state. This will allow a Kalman filter to reconstruct the spacecraft's estimated position and velocity from noisy observations. The spacecraft's "measured" state (mean and covariance) will be maintained using a Kalman or extended Kalman filter, and based on that belief, the POMDP policy will command mode-switching and thrust. We will begin with a simple rule-based strategy—using high-thrust propulsion when far and switching to low-thrust propulsion when near, and performance will be evaluated by fuel use, docking accuracy, and collision-free success rate.

If time allows, we may extend to a chance-constrained model predictive controller (CC-MPC) that adjusts safety margins based on the level of uncertainty. Another optional objective would be adding rigid-body attitude dynamics for a full 6-DOF model. Otherwise, these will be discussed as future work.

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References

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W. H. Clohessy and R. S. Wiltshire. Terminal guidance system for satellite rendezvous. *Journal of the Aerospace Sciences*, 27(9):653–658, 1960.

P. Garrett and M. Isaacson. Autonomous spacecraft servicing: Mode switching in mission design. In *AIAA Space Forum*, page 12–22, 2021.

David A. Scheinman et al. Hybrid propulsion systems for spacecraft: Trends and mission applications. *Acta Astronautica*, 160:456–467, 2019.